Yashwanth Kumar Kotla

Portfolio | linkedin://yashwanthkumarkotla | yashwanthkumarkotla@gmail.com | Austin, Texas, USA

Data Scientist and AI Product Developer with hands-on experience building and deploying end-to-end machine learning systems, predictive analytics platforms, AI applications, and scalable data pipelines. Experienced in statistical modeling, experimental design, explainable AI, NLP, MLOps, model deployment, and product-focused machine learning development. Skilled in Python, XGBoost, Scikit-learn, FastAPI, Docker, MLflow, cloud technologies, and transforming research concepts into production systems.

EXPERIENCE

Aura Duo
Founder/AI Product Developer — Austin, Texas

- Founded and launched Aura Duo, an AI-powered dual-intelligence platform that orchestrates GPT-4 and Gemini to collaboratively generate, critique, and synthesize responses for higher-quality reasoning and reduced hallucinations.
- Designed a multi-agent LLM architecture where independent model outputs undergo cross-evaluation, critique scoring, and response refinement before producing a final synthesized answer.
- Engineered prompt orchestration workflows, evaluation pipelines, and model interaction frameworks that improved response completeness, factual consistency, and reasoning quality across complex user queries.
- Integrated OpenAI and Google Gemini APIs into a scalable production application, handling model routing, response aggregation, error handling, and user interaction workflows.
- Led the end-to-end product lifecycle including system architecture, AI engineering, deployment, testing, user feedback collection, and iterative product improvements.

Freelance Gig — **04/2025 – 09/2025**
Data Scientist — Austin, Texas

- Developed automated ETL pipelines integrating data from multiple business sources, improving data accessibility and reducing manual reporting efforts by approximately 40%.
- Built interactive Tableau and Power BI dashboards to monitor operational KPIs, customer behavior, and campaign performance, enabling faster data-driven decision making.
- Conducted exploratory data analysis, statistical testing, and business performance analysis to identify trends, optimize marketing strategies, and improve campaign ROI.
- Collaborated with stakeholders to translate business requirements into analytical solutions, delivering actionable insights through data visualization and reporting.
- Designed data models and reporting workflows that improved reporting accuracy, consistency, and executive-level visibility into business performance metrics.

PROJECTS

Layoff Radar – Predictive Risk Intelligence Platform — Live app | Github

- Architected and deployed an end-to-end workforce risk intelligence platform that predicts the likelihood of corporate layoffs using financial performance indicators, historical layoff events, and executive sentiment signals extracted from earnings communications.
- Engineered scalable data pipelines integrating 47K+ quarterly financial records from SimFin and 2,300+ layoff events, developing a temporal labeling framework that transformed historical events into predictive training examples for early-warning risk detection.

- Developed and optimized XGBoost-based predictive models using SMOTE, hyperparameter tuning, probability calibration, and business-driven evaluation metrics to identify companies at elevated layoff risk before public announcements.
- Built a financial NLP pipeline leveraging FinBERT to automatically retrieve and analyze CEO/CFO earnings-call transcripts, extracting sentiment and management-confidence signals that were incorporated as predictive features and improved model context beyond traditional financial metrics.
- Implemented SHAP-based explainability to provide transparent feature attribution, enabling users to understand the financial, operational, and sentiment factors driving individual layoff-risk predictions.
- Designed production-grade REST APIs using FastAPI, containerized services with Docker, tracked experiments with MLflow, and deployed interactive analytics applications supporting both single-company and batch risk assessments.
- Developed an explainable decision-support interface that combines predictive risk scores, sentiment analysis, and feature-level explanations to help users evaluate workforce stability and emerging organizational risks.

Retention ROI System

- Architected a multi-layer retention analytics system integrating predictive modeling, causal inference, explainable AI, and LLM-driven decision support to quantify churn risk, intervention effectiveness, and expected retention ROI.
- Engineered end-to-end ML pipelines using Scikit-learn, XGBoost, and LightGBM with calibrated probability estimation, class-imbalance handling, and business-driven threshold optimization for customer retention use cases.
- Applied uplift modeling techniques to estimate treatment effects and identify customer segments most likely to respond positively to retention interventions, improving targeting efficiency beyond traditional churn prediction approaches.
- Automated generation of customer-specific retention strategies by combining SHAP explanations with LLM-based narrative synthesis, producing human-readable recommendations for customer success and marketing teams.
- Deployed containerized REST APIs and interactive business dashboards while implementing model monitoring, drift detection, and experiment tracking to support production-grade machine learning operations.

Insurance Premium Category Predictor Live app | Github

- Developed and deployed an end-to-end insurance risk classification platform that predicts premium categories using demographic, lifestyle, and behavioral risk factors, automating traditional insurance risk assessment workflows.
- Engineered predictive features including BMI, age segmentation, lifestyle risk scoring, and city-tier categorization, transforming raw customer attributes into risk-relevant variables for model inference.
- Built machine learning pipelines using Scikit-learn to classify policyholders into premium risk tiers, enabling real-time underwriting recommendations through a production-ready prediction service.
- Designed and implemented RESTful inference APIs using FastAPI and Pydantic, incorporating schema validation, health monitoring endpoints, and robust request-response workflows for reliable model serving.
- Developed an interactive Streamlit-based decision-support interface that allows users to evaluate insurance risk profiles and receive instant premium category predictions through an intuitive web application.
- Containerized the entire application stack using Docker and published deployment-ready images to Docker Hub, enabling reproducible environments and streamlined deployment across platforms.
- Applied software engineering and MLOps best practices including modular architecture, API-first design, input validation, model lifecycle management, and production deployment workflows.

CORE COMPETENCIES

• Machine Learning • Predictive Modeling • Statistical Analysis • Experimental Design • Feature Engineering • Explainable AI (SHAP) • Natural Language Processing (NLP) • Large Language Models (LLMs) • Risk Modeling • Customer Analytics • Churn Prediction • Uplift Modeling • Classification & Ensemble Methods • Time-Series & Event Prediction • Model Evaluation & Calibration • Causal Inference • Decision Intelligence • Data Visualization • Data Storytelling • Product Analytics • MLOps • Model Deployment • API Development • Cloud-Native Machine Learning

SKILLS

- Programming Languages: Python, SQL
- Machine Learning & AI: Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, SHAP, CausalML, FinBERT, Transformers, OpenAI, Google Gemini, LangChain
- Data Science & Analytics: Statistical Modeling, Hypothesis Testing, A/B Testing, Predictive Analytics, Classification, Feature Engineering, Model Calibration, Uplift Modeling, Customer Segmentation, Explainable AI (XAI), Exploratory Data Analysis (EDA)
- Data Engineering: ETL Pipelines, Data Warehousing, Data Modeling, Data Cleaning, Feature Stores, Data Integration, PostgreSQL, MySQL
- MLOps & Deployment: FastAPI, Streamlit, Docker, Docker Compose, MLflow, Git, GitHub Actions, REST APIs, Model Monitoring, Drift Detection, CI/CD
- Cloud & Infrastructure: AWS (S3, EC2, SageMaker), Google Cloud Platform (GCP), Render
- Visualization & BI: Tableau, Power BI, Plotly, Matplotlib, Seaborn
- Developer Tools: Git, GitHub, Jupyter Notebook, VS Code, Postman

EDUCATION

Webster University **08/2024 – 12/2026**
Master's in Data Science